

SIJIL PELAJARAN MALAYSIA
PEPERIKSAAN RAMALAN / PERCUBAAN TAHUN 2026

MATEMATIK (1449/1)

KERTAS 1 / PAPER 1

1 jam 30 minit / 1 hour 30 minutes

NAMA: _____ ANGKA GILIRAN: _____
TINGKATAN: _____ NO. KAD PENGENALAN: _____

MAKLUMAT UNTUK CALON / INFORMATION FOR CANDIDATES

1. Kertas peperiksaan ini mengandungi **40 soalan**. Jawab **semua** soalan.
2. Tiap-tiap soalan diikuti oleh empat pilihan jawapan, iaitu **A, B, C** dan **D**. Bagi setiap soalan, pilih **satu** jawapan sahaja.
3. Rajah yang mengiringi soalan tidak dilukis mengikut skala kecuali dinyatakan.
4. Anda dibenarkan menggunakan kalkulator saintifik.

RUMUS MATEMATIK / MATHEMATICAL FORMULAE

Perkaitan / Relations:

1. $a^m \times a^n = a^{m+n}$
2. $a^m \div a^n = a^{m-n}$
3. $(a^m)^n = a^{mn}$
4. $I = Prt$
5. $MV = P\left(1 + \frac{r}{n}\right)^{nt}$
6. Kecerunan / Gradient, $m = \frac{y_2 - y_1}{x_2 - x_1}$

Bentuk & Ruang / Shapes & Space:

1. Luas bulatan / Area of circle = πj^2
2. Lilitan bulatan / Circumference = $2\pi j$
3. Luas sektor / Area of sector = $\frac{\theta}{360^\circ} \pi j^2$
4. Panjang lengkok / Arc length = $\frac{\theta}{360^\circ} (2\pi j)$
5. Luas imej / Area of image = $k^2 \times$ Luas objek
6. Min / Mean, $\bar{x} = \frac{\sum x}{N}$

Jawab semua soalan / Answer all questions

1. Ungkapkan $\frac{0.00048 \times 10^{-3}}{1.2 \times 10^5}$ dalam bentuk piawai.

Express $\frac{0.00048 \times 10^{-3}}{1.2 \times 10^5}$ in standard form.

A. 4.0×10^{-12}

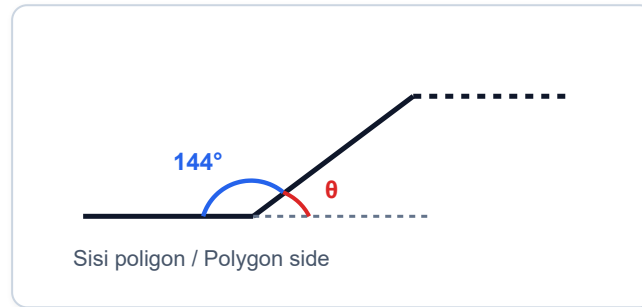
B. 4.0×10^{-11}

C. 4.0×10^{-10}

D. 4.0×10^{-9}

2. Rajah di bawah menunjukkan sebahagian daripada sebuah poligon sekata dengan n sisi. Diberi sudut pedalaman ialah 144° . Cari nilai n .

The diagram below shows part of a regular polygon with n sides. Given that the interior angle is 144° . Find the value of n .



A. 8

B. 9

C. 10

D. 12

3. Garis lurus $2y - 6x = 9$ adalah selari dengan garis lurus $y = (k + 1)x - 4$. Cari nilai k .

The straight line $2y - 6x = 9$ is parallel to the straight line $y = (k + 1)x - 4$. Find the value of k .

A. 2

B. 3

C. 5

D. 6

4. Encik Farid menyimpan RM8,000 dalam akaun simpanan tetap dengan kadar faedah 3.5% setahun yang dikompaun setiap 3 bulan selama 4 tahun. Hitung nilai matang simpanan beliau.

Encik Farid deposits RM8,000 in a fixed deposit account with an interest rate of 3.5% per annum compounded quarterly for 4 years. Calculate the maturity value of his savings.

A. RM9,197.81

B. RM9,190.45

C. RM9,203.20

D. RM9,315.60

5. Diberi $s = \frac{3t+4}{2-t}$. Ungkapkan t dalam sebutan s .

Given $s = \frac{3t+4}{2-t}$. Express t in terms of s .

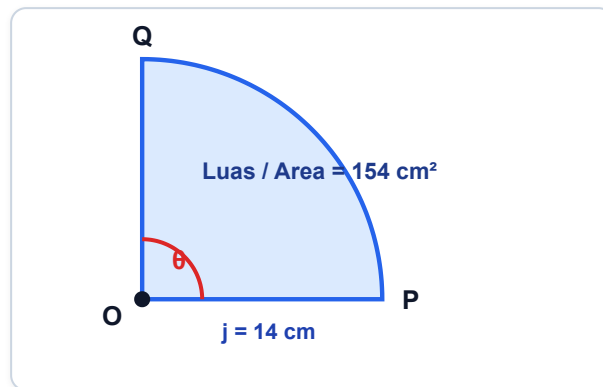
A. $t = \frac{2s-4}{s+3}$

B. $t = \frac{2s+4}{s-3}$

C. $t = \frac{4-2s}{s+3}$

D. $t = \frac{s+3}{2s-4}$

The diagram below shows a sector of a circle OPQ with centre O and radius 14 cm . Given that the area of the sector is 154 cm^2 . Find the value of θ . [Use $\pi = \frac{22}{7}$]



- A. 60°
B. 90°
C. 120°
D. 135°

Which of the following is the locus of a point which is always equidistant from two fixed points P and Q ?

- | | |
|---|--|
| <p>A. Sebuah bulatan berpusat di titik tengah PQ
 <i>A circle centred at the midpoint of PQ</i></p> | <p>B. Garis lurus yang selari dengan garis PQ
 <i>A straight line parallel to line PQ</i></p> |
| <p>C. Pembahagi dua sama serenjang bagi tembereng garis PQ
 <i>Perpendicular bisector of line segment PQ</i></p> | <p>D. Pembahagi dua sama sudut
 <i>Angle bisector</i></p> |

Solve the following linear inequality: $\frac{3-2x}{5} \leq x - 5$.

- A.** $x \geq 4$ **B.** $x \leq 4$
C. $x > -4$ **D.** $x < -4$

Find the coordinates of the maximum point for the quadratic function $f(x) = -2x^2 + 8x - 5$.

- A.** $(2, 3)$
- B.** $(2, -5)$
- C.** $(-2, 3)$
- D.** $(4, 3)$

Express the binary number 110111_2 as a number in base eight.

- A. 57_8
C. 73_8
- B. 67_8
D. 133_8

Calculate the value of $423_5 - 134_5$ and state the answer in base five.

- A. 234_5
C. 289_5
- B. 244_5
D. 344_5

12. Antara pernyataan majmuk berikut, yang manakah bernilai BENAR?

Which of the following compound statements is TRUE?

A. 3 ialah faktor bagi 14 dan $3^2 = 6$
3 is a factor of 14 and $3^2 = 6$

B. $5 > 2$ atau $-3 > -1$
 $5 > 2$ or $-3 > -1$

C. Sebuah heksagon mempunyai 5 sisi dan 2 ialah nombor perdana
A hexagon has 5 sides and 2 is a prime number

D. $4 + 3 = 8$ atau $\sqrt{16} = -4$
 $4 + 3 = 8$ or $\sqrt{16} = -4$

13. Tentukan kontrapositif bagi implikasi berikut: 'Jika x ialah gandaan 6, maka x ialah gandaan 3.'

Determine the contrapositive of the following implication: 'If x is a multiple of 6, then x is a multiple of 3.'

A. Jika x bukan gandaan 6, maka x bukan gandaan 3.
If x is not a multiple of 6, then x is not a multiple of 3.

B. Jika x ialah gandaan 3, maka x ialah gandaan 6.
If x is a multiple of 3, then x is a multiple of 6.

C. Jika x bukan gandaan 3, maka x bukan gandaan 6.
If x is not a multiple of 3, then x is not a multiple of 6.

D. x ialah gandaan 6 jika dan hanya jika x ialah gandaan 3.
 x is a multiple of 6 if and only if x is a multiple of 3.

14. Diberi set semesta $\xi = \{x : 1 \leq x \leq 20, x \text{ ialah integer}\}$, set $P = \{x : x \text{ ialah nombor ganjil}\}$ dan set $Q = \{x : x \text{ ialah nombor perdana}\}$. Cari $n(P \cap Q')$.

Given the universal set $\xi = \{x : 1 \leq x \leq 20, x \text{ is an integer}\}$, set $P = \{x : x \text{ is an odd number}\}$ and set $Q = \{x : x \text{ is a prime number}\}$. Find $n(P \cap Q')$.

A. 2

B. 3

C. 7

D. 8

15. Dalam sebuah kelas dengan 40 orang murid, 25 orang gemar badminton, 18 orang gemar bola sepak dan 5 orang tidak gemar kedua-dua sukan tersebut. Cari bilangan murid yang gemar sukan badminton SAHAJA.

In a class of 40 students, 25 like badminton, 18 like football and 5 do not like both sports. Find the number of students who like badminton ONLY.

A. 8

B. 10

C. 17

D. 20

16. Sebuah graf mempunyai 6 bucu dengan darjah bucu 2, 3, 3, 4, k , 2. Diberi jumlah darjah graf ialah 16. Cari nilai k dan bilangan tepi bagi graf itu.

A graph has 6 vertices with degrees 2, 3, 3, 4, k , 2. Given the sum of degrees of the graph is 16. Find the value of k and the number of edges of the graph.

A. $k = 2$, Bilangan tepi = 8
 $k = 2$, Number of edges = 8

B. $k = 2$, Bilangan tepi = 16
 $k = 2$, Number of edges = 16

C. $k = 4$, Bilangan tepi = 8
 $k = 4$, Number of edges = 8

D. $k = 4$, Bilangan tepi = 16
 $k = 4$, Number of edges = 16

17. Antara pernyataan berikut, yang manakah BUKAN ciri-ciri bagi sebuah pokok dalam teori graf?

Which of the following is NOT a property of a tree in graph theory?

A. Merupakan graf mudah yang bersambung
Is a connected simple graph

B. Bilangan tepi $E = V - 1$
Number of edges $E = V - 1$

C. Mempunyai sekurang-kurangnya satu gelung
Has at least one loop

D. Tidak mengandungi kitaran
Does not contain cycles

22. Suatu set data mempunyai min 14 dan sisihan piawai 3.2. Jika setiap cerapan didarab dengan 2 dan kemudian ditambah dengan 5, cari min dan sisihan piawai yang baharu.

A data set has a mean of 14 and a standard deviation of 3.2. If each data value is multiplied by 2 and then added by 5, find the new mean and new standard deviation.

A. Min = 33, Sisihan piawai = 6.4
Mean = 33, Standard deviation = 6.4

B. Min = 33, Sisihan piawai = 11.4
Mean = 33, Standard deviation = 11.4

C. Min = 28, Sisihan piawai = 6.4
Mean = 28, Standard deviation = 6.4

D. Min = 33, Sisihan piawai = 3.2
Mean = 33, Standard deviation = 3.2

23. Sebuah kotak mengandungi 5 biji guli merah dan 7 biji guli biru. Dua biji guli dipilih secara rawak satu demi satu tanpa pemulangan. Cari kebarangkalian bahawa kedua-dua guli yang dipilih berlainan warna.

A box contains 5 red marbles and 7 blue marbles. Two marbles are selected at random one after another without replacement. Find the probability that both marbles selected are of different colours.

A. $\frac{35}{144}$

B. $\frac{35}{72}$

C. $\frac{35}{66}$

D. $\frac{31}{66}$

24. Pendapatan aktif Encik Chong ialah RM4,500 dan pendapatan pasifnya ialah RM600. Perbelanjaan tetap beliau ialah RM2,200 manakala perbelanjaan tidak tetap ialah RM1,800. Hitung aliran tunai bulanan Encik Chong.

Encik Chong's active income is RM4,500 and his passive income is RM600. His fixed expenses are RM2,200 while variable expenses are RM1,800. Calculate Encik Chong's monthly cash flow.

A. +RM1,100 (Aliran tunai positif)
+RM1,100 (Positive cash flow)

B. +RM500 (Aliran tunai positif)
+RM500 (Positive cash flow)

C. -RM500 (Aliran tunai negatif)
-RM500 (Negative cash flow)

D. +RM1,500 (Aliran tunai positif)
+RM1,500 (Positive cash flow)

25. Diberi bahawa y berubah secara langsung dengan kuasa dua x dan secara songsang dengan punca kuasa dua z . Diberi $y = 12$ apabila $x = 2$ dan $z = 9$. Ungkapkan y dalam sebutan x dan z .

Given that y varies directly as the square of x and inversely as the square root of z . Given $y = 12$ when $x = 2$ and $z = 9$. Express y in terms of x and z .

A. $y = \frac{9x^2}{\sqrt{z}}$

B. $y = \frac{3x^2}{\sqrt{z}}$

C. $y = \frac{36x^2}{\sqrt{z}}$

D. $y = \frac{6x^2}{\sqrt{z}}$

26. Jisim silinder, M , berubah secara langsung dengan tinggi silinder, h , dan kuasa dua jejari, r . Jika tinggi bertambah sebanyak 20% dan jejari berkurang sebanyak 10%, tentukan peratus perubahan jisim silinder itu.

The mass of a cylinder, M , varies directly as its height, h , and the square of its radius, r . If the height increases by 20% and the radius decreases by 10%, determine the percentage change in the cylinder's mass.

A. Berkurang 2.8%
Decreases by 2.8%

B. Meningkat 2.8%
Increases by 2.8%

C. Berkurang 8.0%
Decreases by 8.0%

D. Meningkat 8.0%
Increases by 8.0%

27. Diberi matriks $P = \begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix}$ dan matriks $Q = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$. Hitung PQ .

Given matrix $P = \begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix}$ and matrix $Q = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$. Calculate PQ .

A. $\begin{pmatrix} 16 \\ -18 \end{pmatrix}$

B. $\begin{pmatrix} -4 \\ -18 \end{pmatrix}$

C. $\begin{pmatrix} 16 & -18 \end{pmatrix}$

D. $\begin{pmatrix} 6 & 10 \\ 2 & -20 \end{pmatrix}$

28. Diberi matriks $M = \begin{pmatrix} 2x & 4 \\ 3 & x-1 \end{pmatrix}$ tidak mempunyai songsangan. Cari nilai-nilai yang mungkin bagi x .

Given that matrix $M = \begin{pmatrix} 2x & 4 \\ 3 & x-1 \end{pmatrix}$ does not have an inverse. Find the possible values of x .

A. $x = 3, x = -2$

B. $x = -3, x = 2$

C. $x = 4, x = -3$

D. $x = 6, x = -1$

29. Diberi $\begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10 \\ 12 \end{pmatrix}$. Cari nilai x dan nilai y .

Given $\begin{pmatrix} 4 & -1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10 \\ 12 \end{pmatrix}$. Find the value of x and of y .

A. $x = 3, y = 2$

B. $x = 2, y = 3$

C. $x = 4, y = 1$

D. $x = 1, y = -6$

30. Puan Lily ingin membeli polisi insurans hayat bernilai RM250,000 dengan kadar premium tahunan RM2.45 bagi setiap RM1,000 nilai muka. Beliau turut menambah rider penyakit kritikal bernilai RM100,000 dengan kadar RM1.80 bagi setiap RM1,000. Hitung jumlah premium tahunan beliau.

Puan Lily wants to buy a life insurance policy worth RM250,000 with an annual premium rate of RM2.45 per RM1,000 face value. She also adds a critical illness rider worth RM100,000 with a rate of RM1.80 per RM1,000. Calculate her total annual premium.

A. RM612.50

B. RM792.50

C. RM850.00

D. RM967.50

31. Encik Kumar mempunyai polisi insurans perubatan dengan peruntukan deduktibel sebanyak RM500 dan fasal penyertaan ko-insurans 80/20. Jika kos rawatan hospital beliau ialah RM12,500, hitung jumlah kos yang perlu ditanggung sendiri oleh Encik Kumar.

Encik Kumar has a medical insurance policy with a deductible provision of RM500 and an 80/20 co-insurance clause. If his hospital treatment cost is RM12,500, calculate the total cost borne by Encik Kumar himself.

A. RM2,400

B. RM2,900

C. RM9,600

D. RM10,100

32. Nilai sewa tahunan bagi rumah Encik Halim dianggarkan RM18,000 dan kadar cukai pintu ialah 4%. Hitung jumlah cukai pintu yang perlu dibayar bagi setiap setengah tahun.

The annual rental value of Encik Halim's house is estimated at RM18,000 and the property assessment tax rate is 4%. Calculate the assessment tax payable for each half-year.

A. RM180

B. RM360

C. RM720

D. RM1,440

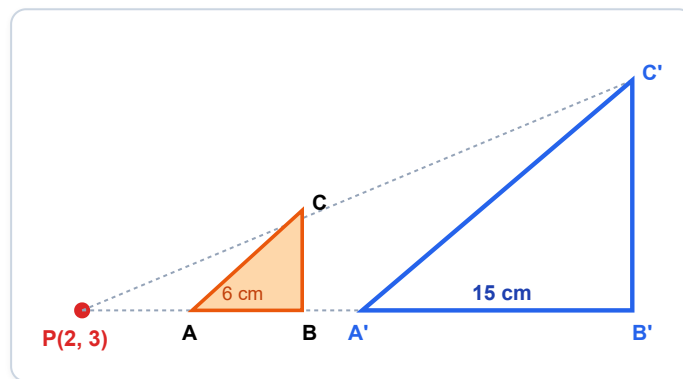
33. Pendapatan tahunan Puan Aina ialah RM68,000. Beliau telah menderma RM1,500 kepada badan kebajikan yang diluluskan dan jumlah pelepasan cukai yang dituntut ialah RM16,500. Diberi jadual kadar cukai: RM35,000 pertama dikenakan RM600 dan kadar seterusnya ialah 8%. Hitung cukai pendapatan yang perlu dibayar oleh Puan Aina.

Puan Aina's annual income is RM68,000. She donated RM1,500 to an approved charity and claimed a total tax relief of RM16,500. Given the tax schedule: the first RM35,000 is taxed at RM600 and the subsequent rate is 8%. Calculate the income tax payable by Puan Aina.

- A. RM1,600
B. RM1,800
C. RM2,100
D. RM2,640

34. Rajah di bawah menunjukkan pembesaran segi tiga ABC kepada segi tiga $A'B'C'$ dengan pusat $P(2, 3)$. Diberi panjang $AB = 6$ cm dan $A'B' = 15$ cm. Tentukan faktor skala pembesaran itu.

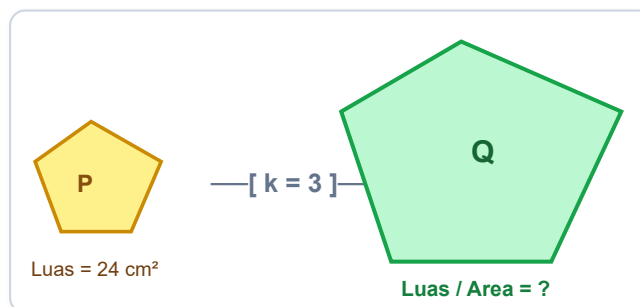
The diagram below shows the enlargement of triangle ABC to triangle $A'B'C'$ with centre $P(2, 3)$. Given the length of $AB = 6$ cm and $A'B' = 15$ cm. Determine the scale factor of the enlargement.



- A. $\frac{2}{5}$
B. $\frac{5}{2}$
C. $-\frac{5}{2}$
D. $\frac{3}{5}$

35. Rajah di bawah menunjukkan bentuk P dipetakan kepada bentuk Q di bawah suatu pembesaran dengan faktor skala $k = 3$. Diberi luas bentuk P ialah 24 cm^2 . Hitung luas bentuk Q .

The diagram below shows shape P mapped onto shape Q under an enlargement with scale factor $k = 3$. Given that the area of shape P is 24 cm^2 . Calculate the area of shape Q .



- A. 72
B. 144
C. 216
D. 240

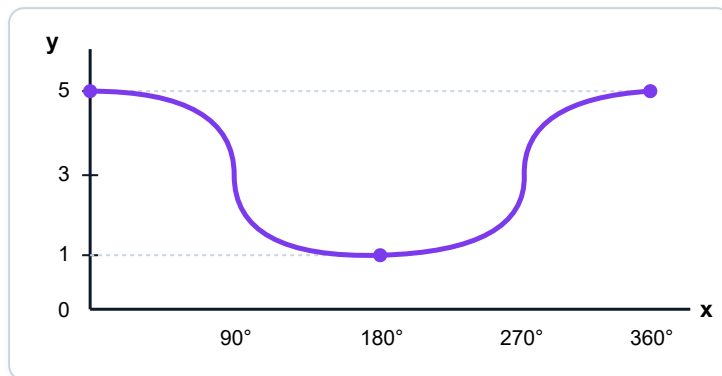
36. Diberi $\tan \theta = -0.75$ dan θ terletak dalam Sukuan II ($90^\circ < \theta < 180^\circ$). Cari nilai bagi $\cos \theta$.

Given that $\tan \theta = -0.75$ and θ lies in Quadrant II ($90^\circ < \theta < 180^\circ$). Find the value of $\cos \theta$.

- A. 0.6
B. -0.6
C. 0.8
D. -0.8

37. Rajah di bawah menunjukkan graf fungsi trigonometri $y = a \cos(bx) + c$ untuk $0^\circ \leq x \leq 360^\circ$. Cari nilai a , nilai b dan nilai c .

The diagram below shows the graph of a trigonometric function $y = a \cos(bx) + c$ for $0^\circ \leq x \leq 360^\circ$. Find the values of a , b and c .



A. $a = 2, b = 1, c = 3$

B. $a = 3, b = 1, c = 2$

C. $a = 2, b = 2, c = 3$

D. $a = 4, b = 1, c = 1$

38. Jadual kekerapan menunjukkan masa ulangkaji (jam) bagi sekumpulan murid: 11-20 (4), 21-30 (9), 31-40 (18), 41-50 (12), 51-60 (7). Tentukan kelas mod dan titik tengah bagi kelas mod itu.

The frequency table shows the revision time (hours) of a group of students: 11-20 (4), 21-30 (9), 31-40 (18), 41-50 (12), 51-60 (7). Determine the modal class and its midpoint.

A. 21-30, 25.5

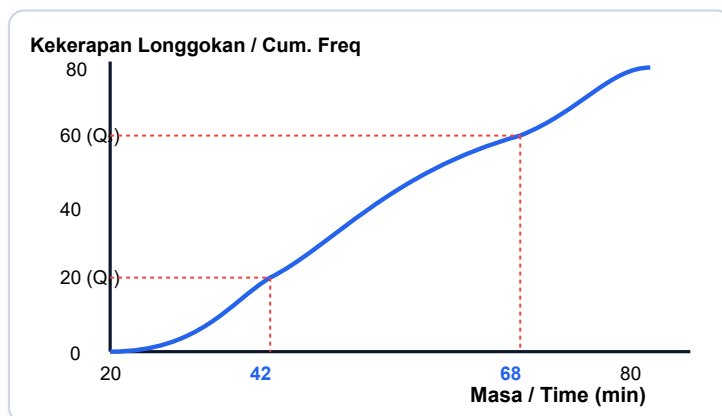
B. 31-40, 35.5

C. 41-50, 45.5

D. 31-40, 35.0

39. Rajah di bawah menunjukkan sebuah ogif bagi masa menunggu pelanggan di sebuah klinik. Diberi $Q_1 = 42$ minit dan $Q_3 = 68$ minit. Hitung julat antara kuartil bagi data tersebut.

The diagram below shows an ogive of customer waiting times at a clinic. Given $Q_1 = 42$ minutes and $Q_3 = 68$ minutes. Calculate the interquartile range of the data.



A. 20

B. 26

C. 40

D. 48

40. Antara berikut, yang manakah menunjukkan urutan yang betul bagi peringkat-peringkat dalam kitaran pemodelan matematik?

Which of the following shows the correct sequence of stages in the mathematical modeling cycle?

A. Mengenal pasti masalah → Membuat andaian →
Mengaplikasi matematik → Menentusahkan penyelesaian
→ Memurnikan model → Melaporkan dapatan

*Identifying the problem → Making assumptions → Applying
mathematics → Verifying solutions → Refining the model →
Reporting findings*

C. Mengenal pasti masalah → Melaporkan dapatan →
Memurnikan model → Membuat andaian

*Identifying the problem → Reporting findings → Refining the
model → Making assumptions*

B. Membuat andaian → Mengenal pasti masalah →
Mengaplikasi matematik → Melaporkan dapatan

*Making assumptions → Identifying the problem → Applying
mathematics → Reporting findings*

D. Mengenal pasti masalah → Mengaplikasi matematik →
Membuat andaian → Melaporkan dapatan

*Identifying the problem → Applying mathematics → Making
assumptions → Reporting findings*

KERTAS PEPERIKSAAN TAMAT / END OF QUESTION PAPER